nitrogen nucleus is represented by ${}^{15}_{7}N$. It implies that nitrogen nucleus contains 15 nucleons of which 7 are protons ($Z=7$) and 8 are neutrons ($N=A-Z=8$). Note that once the element is specified, the value of Z is known and subscript Z is sometimes omitted. For example, nitrogen nucleus is simply denoted as ${}^{15}N$ and we call it as 'nitrogen fifteen'.

Since the nucleus is made up of positively charged protons and electrically neutral neutrons, the overall charge of the nucleus is positive and it has the value $+Ze$. But the atom is electrically neutral which implies that the number of electrons in the atom is equal to the number of protons in the nucleus.

9.4.2 Isotopes, isobars, and isotones

Isotopes:

In nature, there are atoms of a particular element whose nuclei have same number of protons but different number of neutrons. These kinds of atoms are called isotopes. In other words, **isotopes are atoms of the same element having same atomic number *Z*, but different mass number *A***. For example, hydrogen has three isotopes and they are represented as ${}^{1}_{1}H$ (hydrogen), ${}^{2}_{1}H$ (deuterium),and ${}^{3}_{1}H$ (tritium). Note that all the three nuclei have one proton and, hydrogen has no neutron, deuterium has 1 neutron and tritium has 2 neutrons.

The number of isotopes for the particular element and their relative abundances (percentage) vary with each element. For example, carbon has four main isotopes: ${}^{11}_{6}C$, ${}^{12}_{6}C$, ${}^{13}_{6}C$ and ${}^{14}_{6}C$. But in nature, the percentage of ${}^{12}_{6}C$ is approximately 98.9%, that of ${}^{13}_{6}C$ is 1.1% and that of ${}^{14}_{6}C$ is 0.0001%. The other carbon isotope ${}^{11}_{6}C$, does not occur naturally and it can be produced only in nuclear reactions in the laboratory or by cosmic rays.

Since the chemical properties of any atom are determined only by electrons, the isotopes of any element have same electronic structure and same chemical properties. So the isotopes of the same element are placed in the same location in the periodic table.

Isobars:

Isobars are the atoms of different elements having the same mass number *A*, but different atomic number *Z*. In other words, isobars are the atoms of different chemical elements which have same number of nucleons. For example ${}^{40}_{16}S$, ${}^{40}_{17}Cl$, ${}^{40}_{18}Ar$, ${}^{40}_{19}K$ and ${}^{40}_{20}Ca$ are isobars having same mass number 40 but different atomic numbers. Unlike isotopes, isobars are chemically different elements. They have different physical and chemical properties.

Isotones:

Isotones are the atoms of different elements having same number of neutrons. ${}^{12}_{5}B$ and ${}^{13}_{6}C$ are examples of isotones with 7 neutrons each.

9.4.3 Atomic and nuclear masses

The mass of nuclei is very small (about 10^{-25} kg or less). Therefore, it is more convenient to express it in terms of another unit namely, the *atomic mass unit (u)*. **One atomic mass unit (*u*) is defined as the (1/12)th of the mass of the isotope of carbon** ${}^{12}_{6}C$ which is more abundant in naturally occurring isotope of carbon.

In other words

$$1\,u = \frac{\text{mass of } {}^{12}_{6}C \text{ atom}}{12} = \frac{1.9926\times10^{-26}}{12}$$
$$= 1.660\times10^{-27}\,\text{kg}$$

UNIT-9(XII-Physics_Vol-2)_New.indd 164 21-07-2020 15:49:06